

GUARULHOS, Brazil

WMO: 830750

Lat: 23.432S

Lon: 46.470W

Elev: 750

StdP: 92.64

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	9.2	4.9	5.9	16.2	6.8	6.7	15.5	7.8	21.4	7.0	20.4	1.5	80	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.4	32.9	21.6	31.8	21.5	30.8	21.4	24.5	28.5	23.8	28.0	23.2	27.4	4.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	19.8	25.3	22.9	19.3	25.2	22.1	18.4	24.6	78.4	28.7	75.8	28.9	73.4	26.6	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	6.2	5.5	4.7	34.7	1.8	1.0	3.4	35.4	2.3	36.0	1.3	36.5	0.0	37.2		
			4.5	25.7	1.9	1.4	3.1	26.7	2.0	27.5	0.9	28.3	-0.4	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	23.7	24.0	23.1	21.5	18.5	17.5	17.1	18.3	19.6	21.1	21.5	23.1
	DBStd	3.48	2.32	2.04	2.03	2.30	2.45	2.52	2.71	2.85	3.11	3.02	2.70	2.29
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD18.3	206	1	0	1	4	28	44	54	35	22	10	6	1
	CDD10.0	3914	424	392	406	344	264	224	221	258	289	343	343	407
	CDD18.3	1077	166	158	147	98	34	18	16	34	61	96	100	149
	CDH23.3	6491	950	969	742	490	164	119	164	358	500	609	560	864
CDH26.7	2029	329	346	222	112	19	6	15	89	191	239	172	288	
Wind	WSAvg	2.6	2.7	2.6	2.5	2.4	2.4	2.5	2.6	2.9	2.9	2.9	2.9	2.8
Precipitation	PrecAvg	1724	295	227	206	105	80	70	68	44	103	137	167	222
	PrecMax	2404	598	411	399	193	189	266	187	98	225	217	310	398
	PrecMin	1291	166	108	89	23	21	9	17	14	32	44	75	87
	PrecStd	255	101	75	60	41	33	55	45	22	52	42	56	67
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	33.2	32.1	30.9	28.2	27.2	28.2	31.0	33.2	34.0	32.2	33.0
		MCWB	23.2	22.4	22.4	21.7	19.8	18.2	17.5	18.3	19.6	20.7	21.1	22.7
	2%	DB	31.9	32.0	30.8	29.2	27.0	26.1	26.9	29.0	31.1	31.9	30.2	31.2
		MCWB	22.8	22.4	22.1	21.2	19.3	17.9	16.9	17.5	18.4	20.3	21.0	22.1
	5%	DB	30.1	30.8	29.1	28.0	25.2	24.9	25.2	27.2	29.0	29.8	28.9	30.0
		MCWB	22.6	22.2	22.1	20.9	18.5	17.4	16.8	17.0	18.3	20.4	21.1	21.9
	10%	DB	28.2	29.0	27.8	26.2	23.2	23.0	23.6	25.2	26.2	26.9	26.9	28.1
		MCWB	22.4	22.4	21.9	20.4	18.1	17.2	16.5	16.8	18.3	20.1	20.6	21.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	25.3	24.5	23.7	21.6	20.5	19.2	20.2	21.8	23.1	23.6	25.0
		MCDB	29.5	28.9	28.7	27.3	25.5	24.8	24.4	26.4	28.9	29.7	28.7	28.9
	2%	WB	24.3	24.2	23.6	22.7	20.6	19.2	18.5	19.1	20.6	22.0	22.6	23.9
		MCDB	28.4	28.0	27.8	26.6	24.2	23.0	23.6	25.1	26.4	27.6	27.5	28.5
	5%	WB	23.6	23.6	23.0	22.0	19.8	18.5	17.7	18.3	19.7	21.2	21.9	23.1
		MCDB	27.6	27.5	27.0	25.6	23.2	22.2	22.6	24.1	25.1	26.3	26.5	27.6
	10%	WB	23.0	23.0	22.3	21.3	18.9	17.7	17.1	17.5	18.9	20.5	21.2	22.4
		MCDB	26.9	26.9	25.8	24.6	21.9	21.3	21.6	22.6	23.7	24.9	25.0	26.4
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.4	8.9	9.2	9.0	10.0	10.7	11.7	10.5	9.8	9.2	9.4
		MCDBR	11.6	12.2	11.2	11.3	11.7	12.8	14.3	15.9	16.0	14.6	12.9	12.2
		MCWBR	4.3	4.2	4.2	4.4	4.8	5.5	5.9	5.9	5.5	5.0	4.9	4.7
	5% WB	MCDBR	10.1	10.4	9.9	9.8	10.0	10.6	11.8	12.8	12.3	11.6	10.6	10.4
		MCWBR	4.2	4.3	4.2	4.2	4.5	5.3	5.4	5.5	5.1	4.8	4.6	4.6
Clear-Sky Solar Irradiance	taub	0.395	0.394	0.383	0.367	0.348	0.349	0.359	0.394	0.486	0.449	0.421	0.403	
	taud	2.459	2.444	2.479	2.495	2.522	2.502	2.457	2.329	2.057	2.232	2.324	2.410	
	Ebn at Noon	949	937	920	892	868	845	847	849	810	876	919	943	
	Edh at Noon	120	120	111	100	90	88	95	117	166	147	137	127	
All-Sky Solar Radiation	RadAvg	5.19	5.50	4.88	4.42	3.56	3.28	3.45	4.17	4.43	4.95	5.21	5.46	
	RadStd	0.75	0.49	0.48	0.37	0.27	0.32	0.36	0.36	0.56	0.56	0.30	0.39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.30	-0.40	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page